FABRICATION SPECIFICATION
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Project: USB3 Fiber Link

GENERAL
- Layers: 4 (TOP / GND / PWR / BOTTOM)
- Material: FR4, Standard Tg (150-160C)
- Finished Thickness: 62 mil (1.575mm) +/- 10%
- Surface Finish: ENIG (1u")
- Solder Mask: Green, both sides
- Silkscreen: White, both sides
- IPC Class: 2

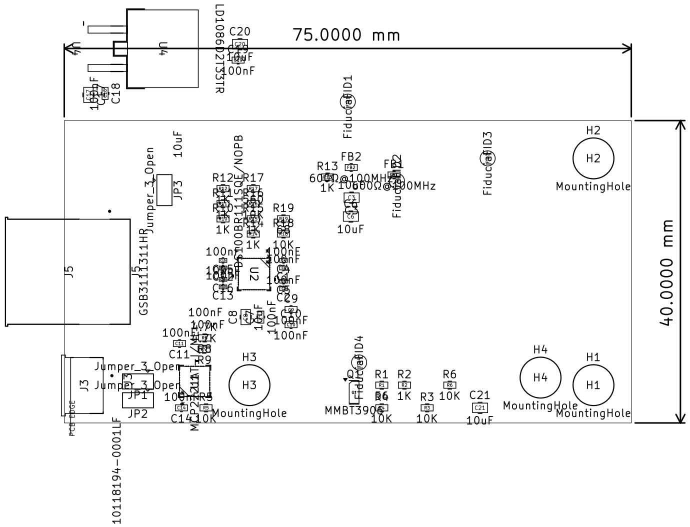
STACKUP
- F.Cu: 1 oz (0.035mm) - Signal
- Prepreg: 0.21mm (7628, Er=4.5)
- In1.Cu: 0.5 oz (0.0175mm) - GND Plane
- Core: 1.03mm (Er=4.5)
- In2.Cu: 0.5 oz (0.0175mm) - PWR Plane
- Prepreg: 0.21mm (7628, Er=4.5)
- B.Cu: 1 oz (0.035mm) - Signal

CONTROLLED IMPEDANCE
- USB3 SuperSpeed: 100 ohm diff, 50 ohm SE
- SFP+ CML: 100 ohm diff, 50 ohm SE
- USB2 High-Speed: 90 ohm diff, 45 ohm SE
- Tolerance: +/-10% (+/-5 ohm if <= 50 ohm)
- Verify trace width with fab stackup calculator

MINIMUM FEATURES (JLCPCB 4-layer 1oz)
- Track width: 0.10mm (4 mil)
- Track spacing: 0.10mm (4 mil)
- Via drill/dia: 0.30mm / 0.50mm
- Annular ring: 0.10mm
- Hole-to-hole: 0.25mm
- Copper-to-edge: 0.50mm
- Solder mask bridge: 0.10mm
- Mask-to-trace: 0.09mm

VIA PLUGGING
- Fill vias with soldermask (opaque)
- Max via diameter for plugging: 0.5mm

NOTES
1. All dimensions in mm unless noted
2. Panelize per fab requirements
3. No blind/buried vias required

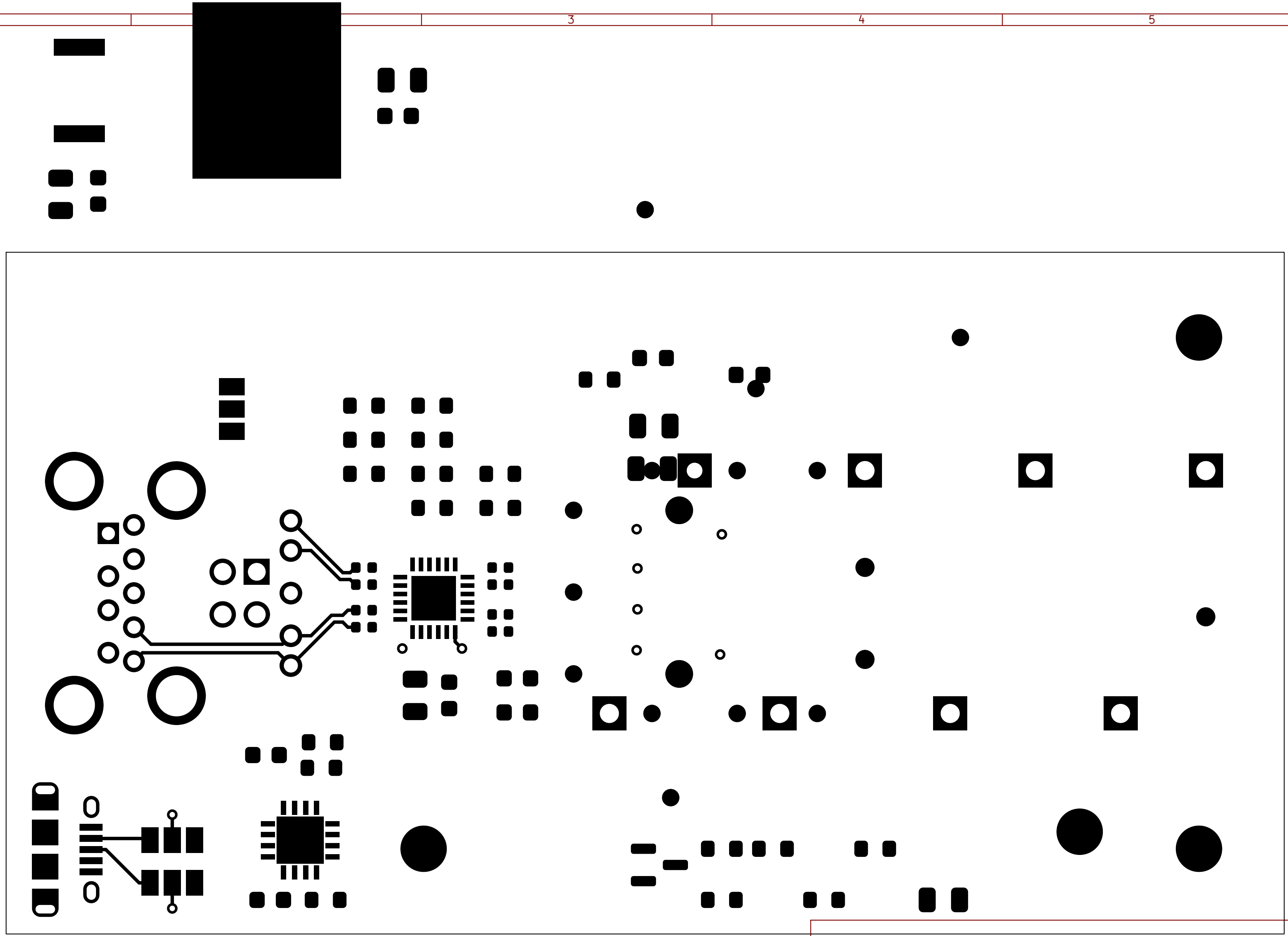


USB 3.0 to SFP+ Board

LAYER: Overview

Repo: github.com/ringof/usb3-fiber

Designer: David Goncalves	Commit: local	OSH Licence: CERN-OHL-P-2.0
Size: A4	Date: 2026-03-13	Rev: DEV
	File: usb3_fiber.kicad_pcb	Page 1 of 11



USB 3.0 to SFP+ Board

LAYER: F.Cu

Repo: github.com/ringof/usb3-fiber

Designer: David Goncalves	Commit: local	OSH Licence: CERN-OHL-P-2.0
Size: A4	Date: 2026-03-13	**Rev: DEV**
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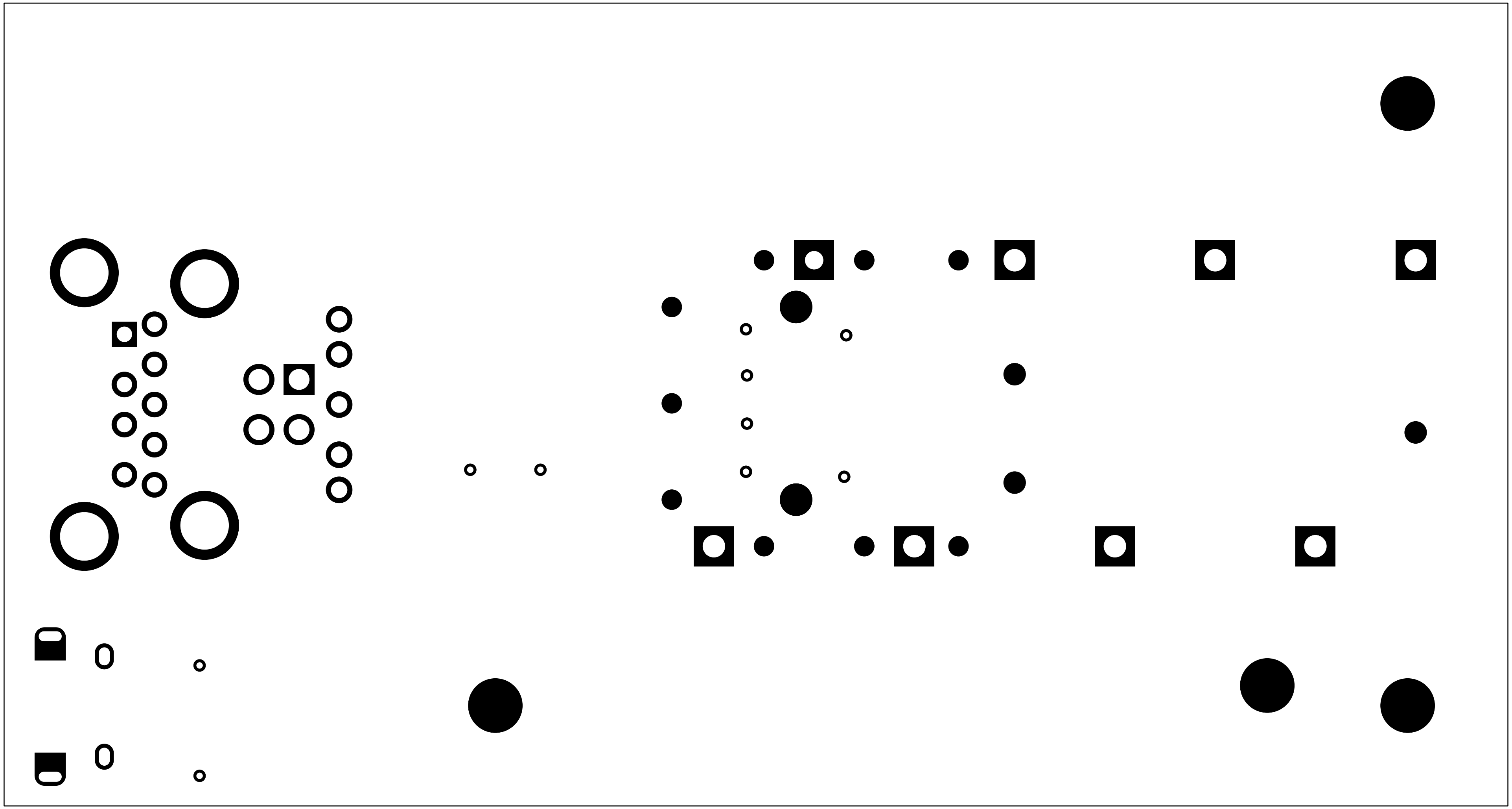
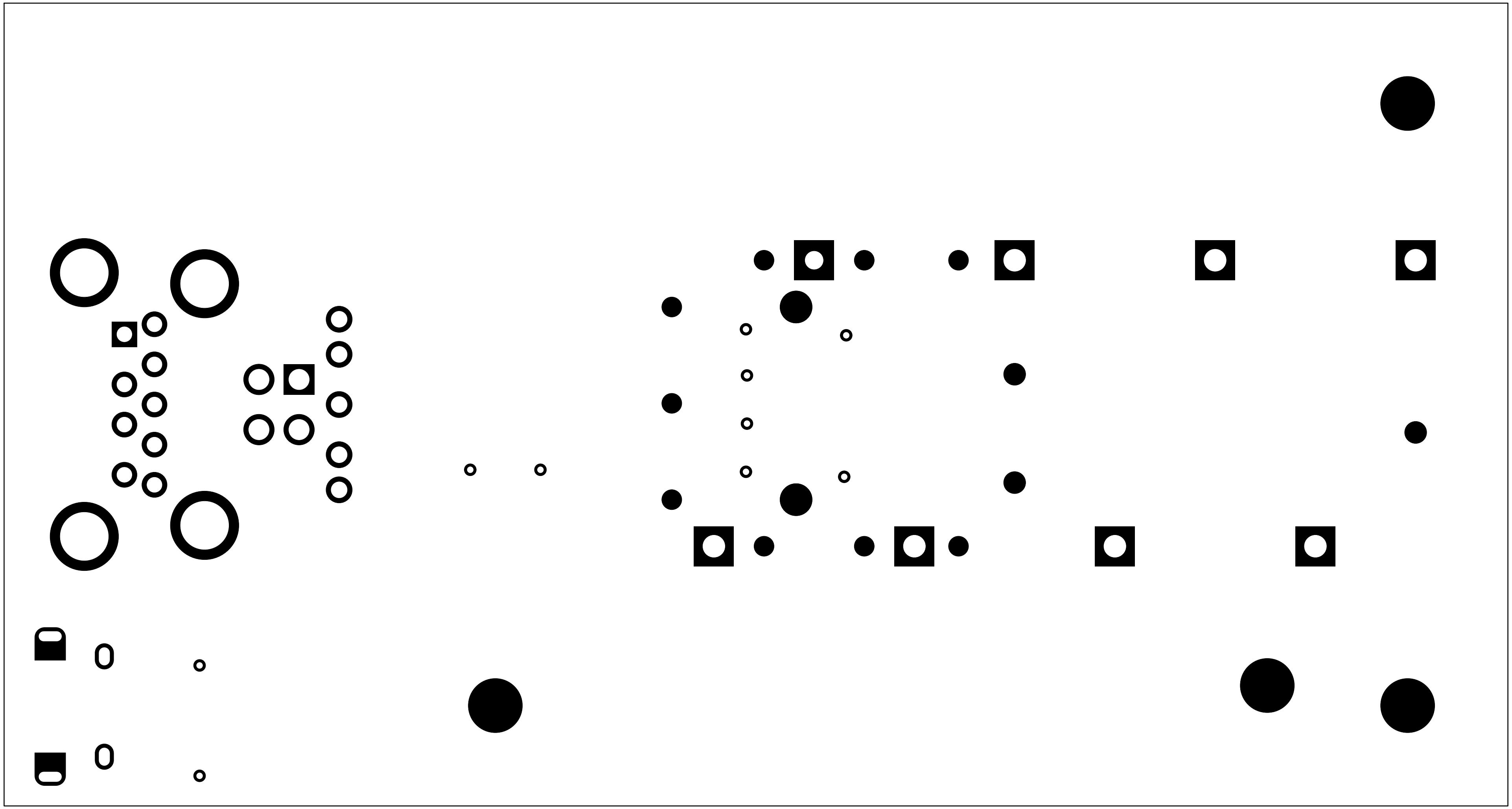


USB 3.0 to SFP+ Board

LAYER: In1.Cu

Repo: github.com/ringof/usb3-fiber

Designer: David Goncalves	Commit: local	OSH Licence: CERN-OHL-P-2.0
Size: A4	Date: 2026-03-13	**Rev: DEV**
	File: usb3_fiber.kicad_pcb	**Page 3 of 11**

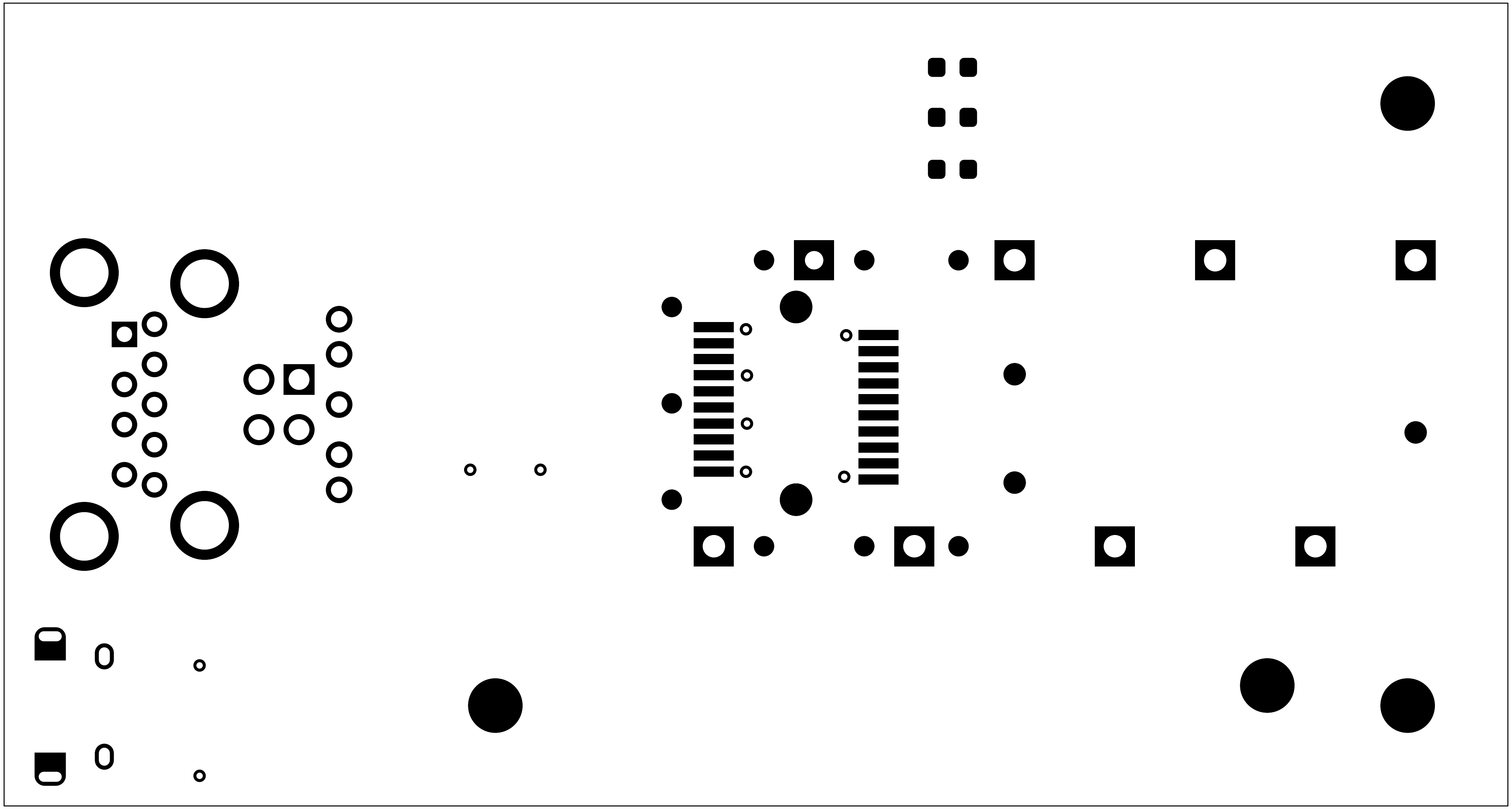


USB 3.0 to SFP+ Board

LAYER: In2.Cu

Repo: github.com/ringof/usb3-fiber

Designer: David Goncalves	Commit: local	OSH Licence: CERN-OHL-P-2.0
Size: A4	Date: 2026-03-13	**Rev: DEV**
	File: usb3_fiber.kicad_pcb	**Page 4 of 11**

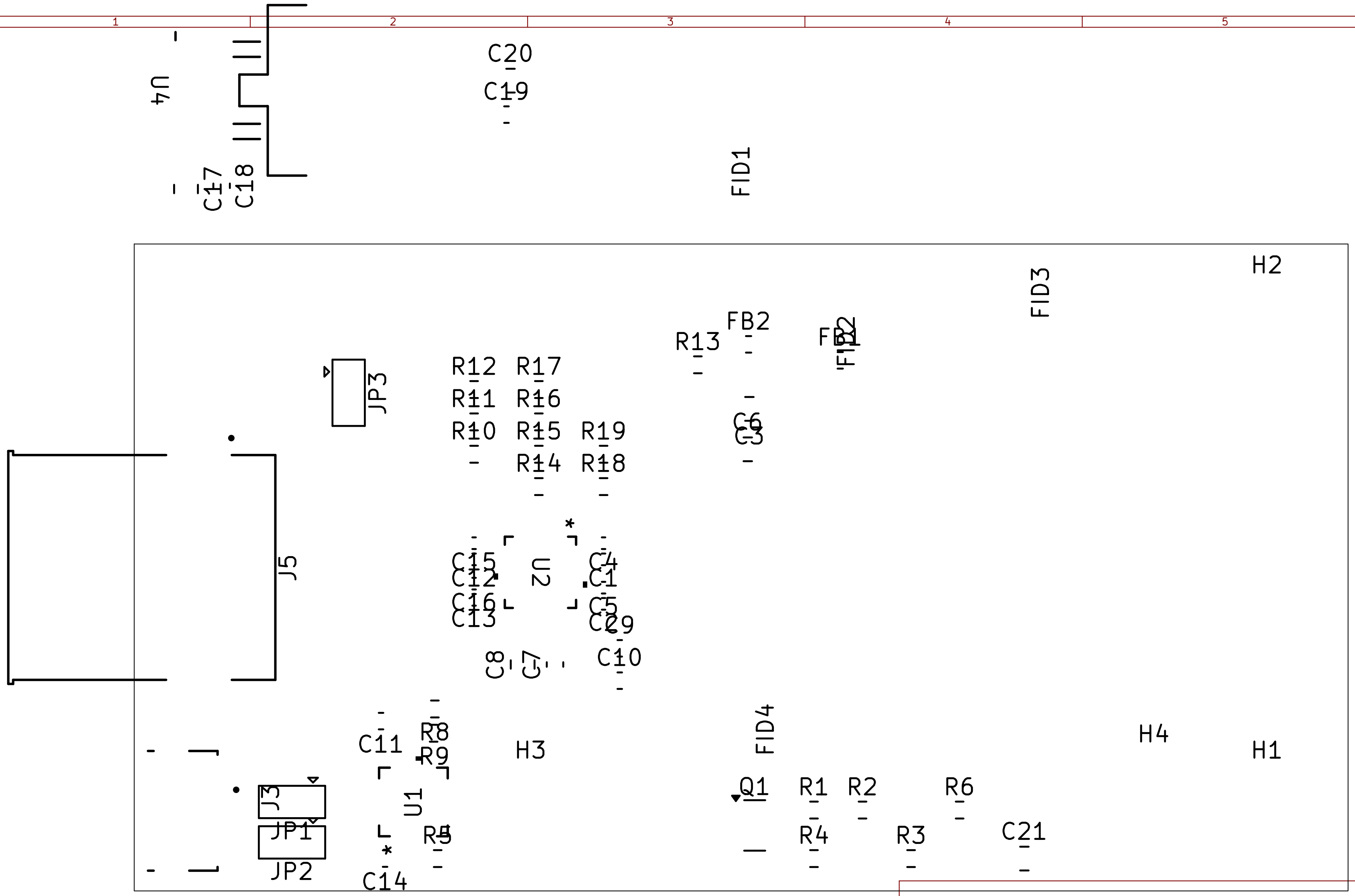


USB 3.0 to SFP+ Board

LAYER: B.Cu

Repo: github.com/ringof/usb3-fiber

Designer: David Goncalves	Commit: local	OSH Licence: CERN-OHL-P-2.0
Size: A4	Date: 2026-03-13	**Rev: DEV**
	File: usb3_fiber.kicad_pcb	**Page 5 of 11**



USB 3.0 to SFP+ Board

LAYER: F.Silkscreen

Repo: github.com/ringof/usb3-fiber

Designer: David Goncalves	Commit: local	OSH Licence: CERN-OHL-P-2.0
Size: A4	Date: 2026-03-13	**Rev: DEV**
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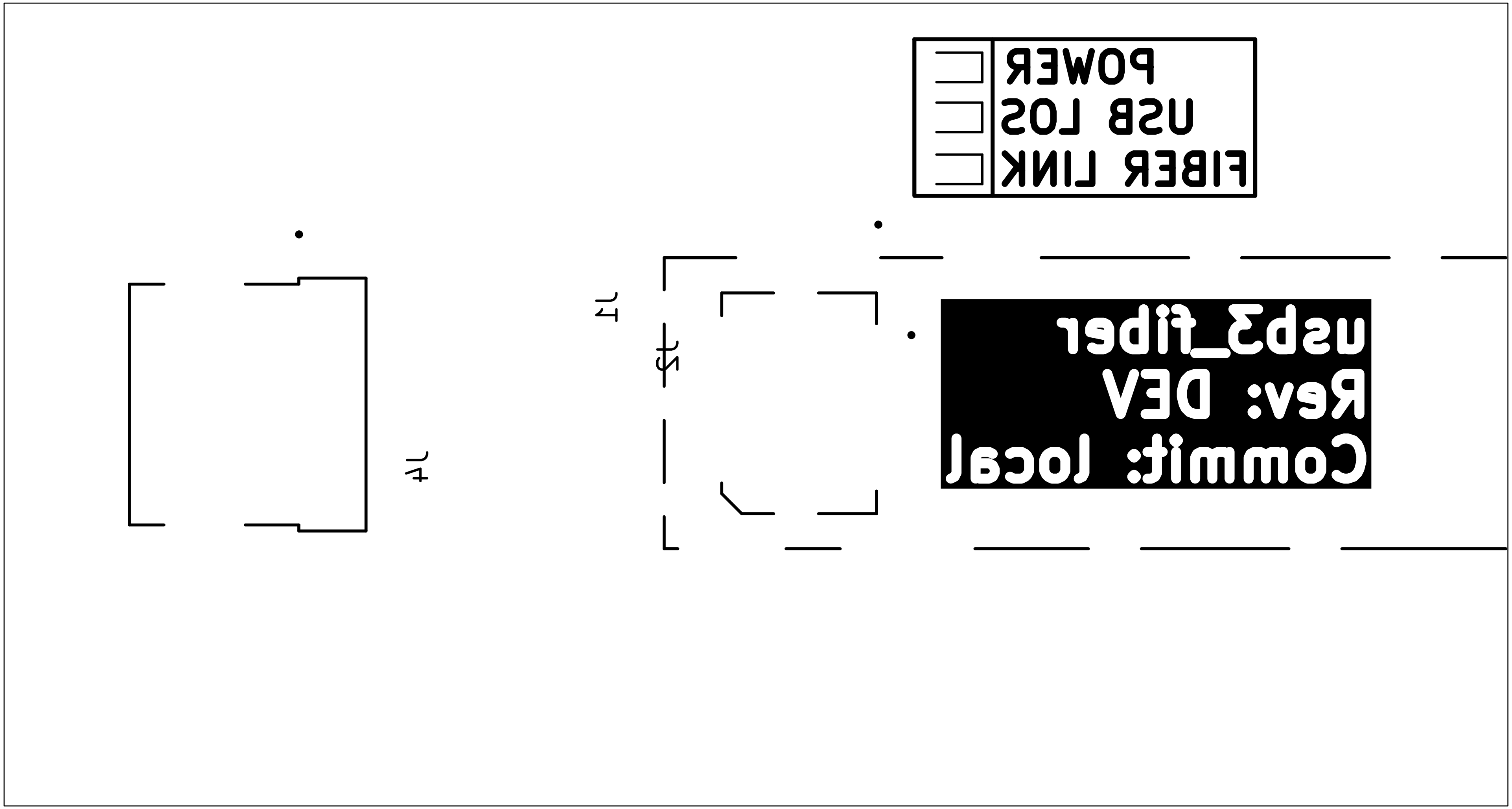

USB 3.0 to SFP+ Board

LAYER: B.Silkscreen

Repo: github.com/ringof/usb3-fiber

Designer: David Goncalves	Commit: local	OSH Licence: CERN-OHL-P-2.0
Size: A4	Date: 2026-03-13	**Rev: DEV**
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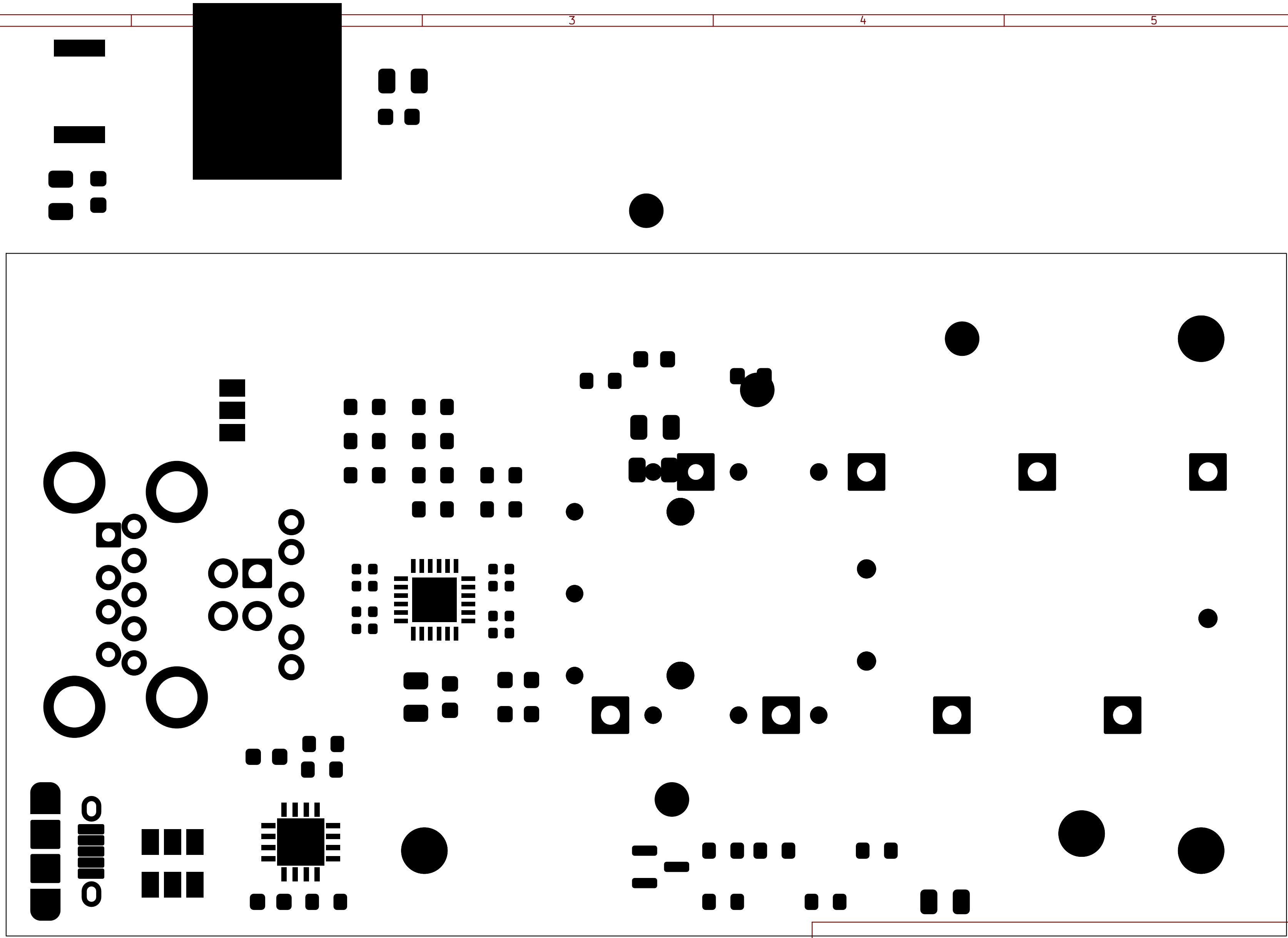

USB 3.0 to SFP+ Board

LAYER: F.Mask

Repo: github.com/ringof/usb3-fiber

Designer: David Goncalves	Commit: local	OSH Licence: CERN-OHL-P-2.0
Size: A4	Date: 2026-03-13	**Rev: DEV**
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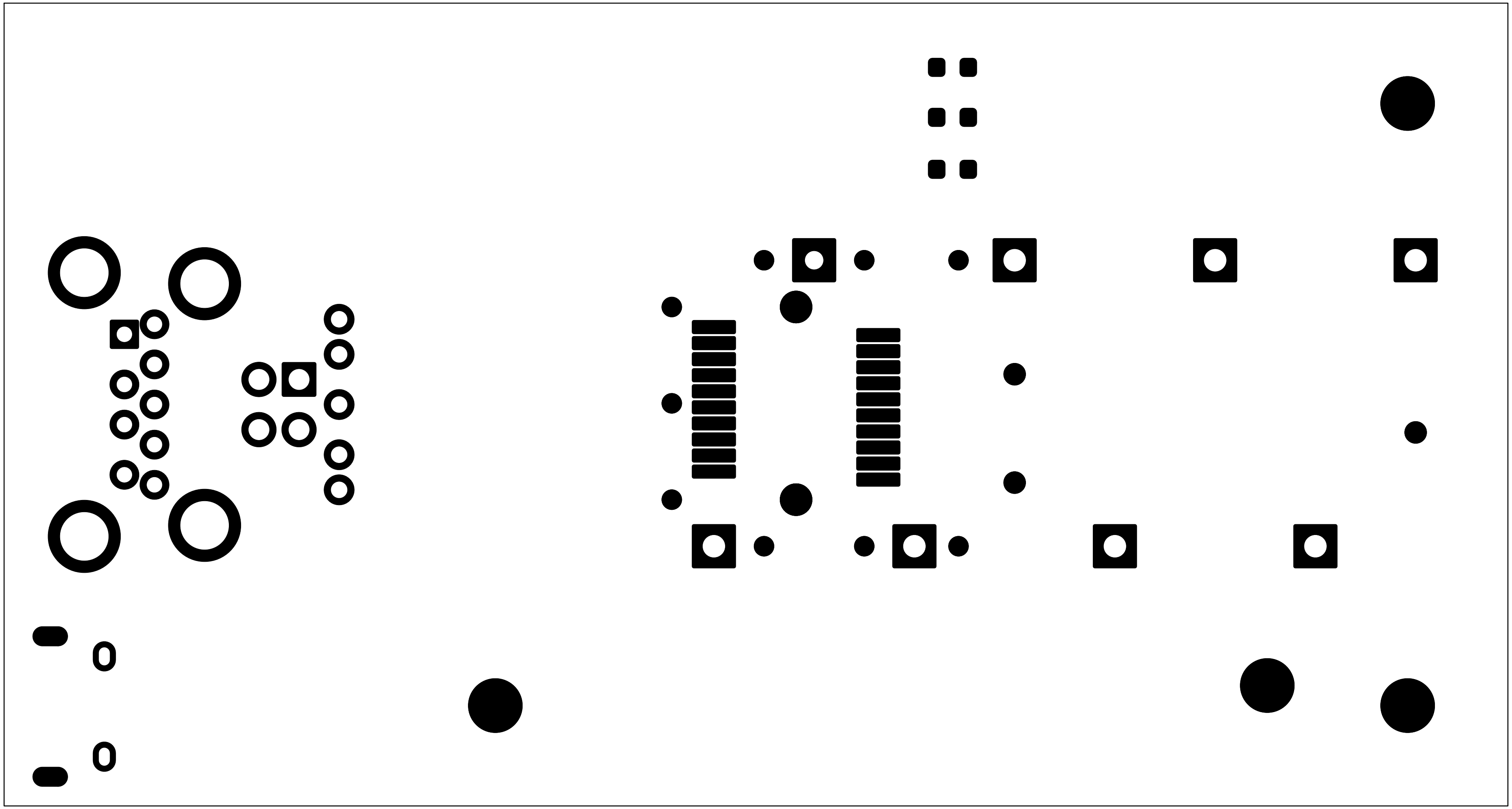

USB 3.0 to SFP+ Board

LAYER: B.Mask

Repo: github.com/ringof/usb3-fiber

Designer: David Goncalves	Commit: local	OSH Licence: CERN-OHL-P-2.0
Size: A4	Date: 2026-03-13	**Rev: DEV**
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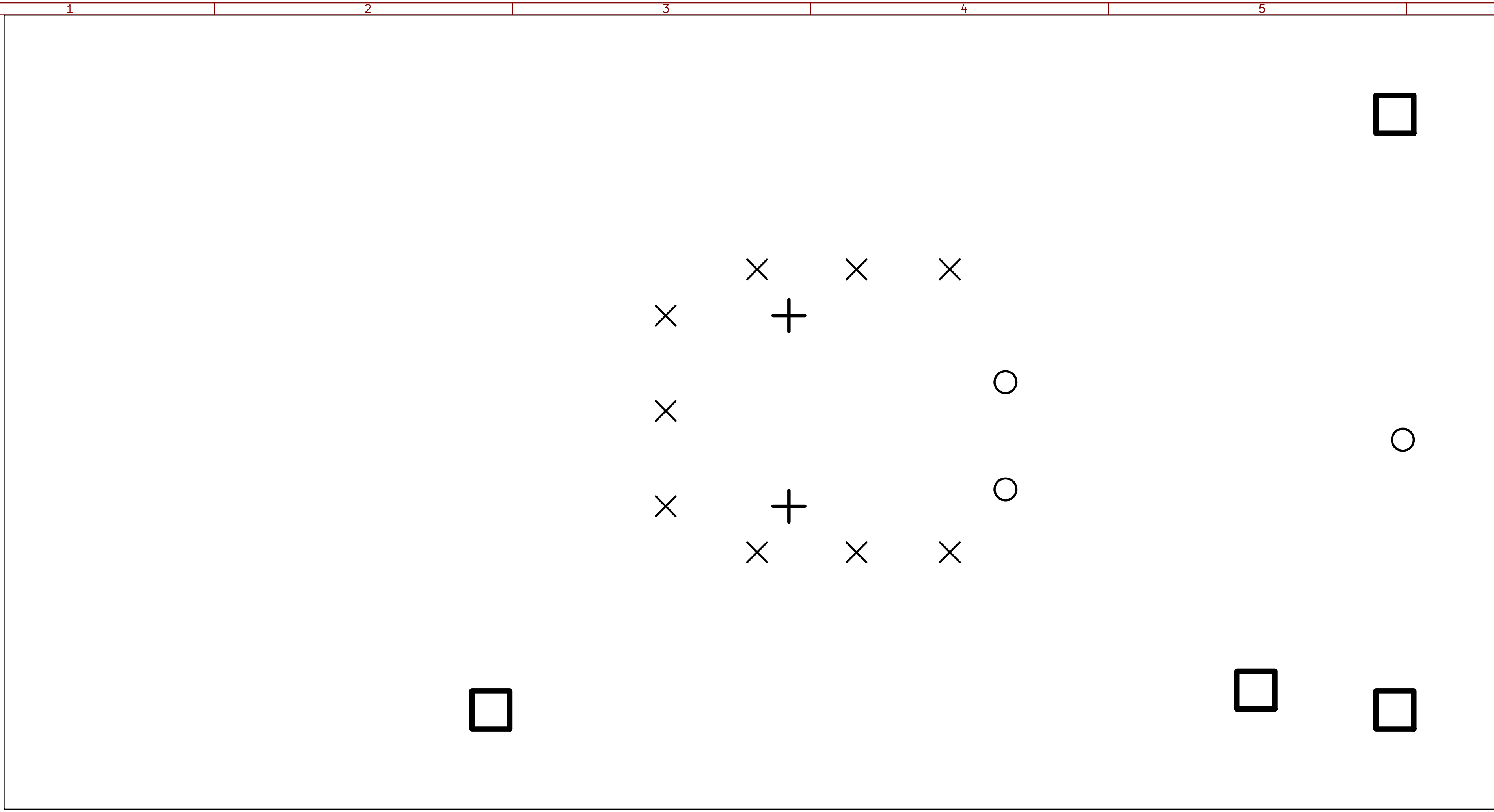

Drill Map:

× 1.000mm / 0.0394" (9 holes) (not plated)
○ 1.100mm / 0.0433" (3 holes) (not plated)
+ 1.600mm / 0.0630" (4 holes) (not plated)
□ 2.700mm / 0.1063" (4 holes) (not plated)

USB 3.0 to SFP+ Board

LAYER: Drill Map (NPTH)

Repo: github.com/ringof/usb3-fiber

Designer: David Goncalves	Commit: local	OSH Licence: CERN-OHL-P-2.0
Size: A4	Date: 2026-03-13	**Rev: DEV**
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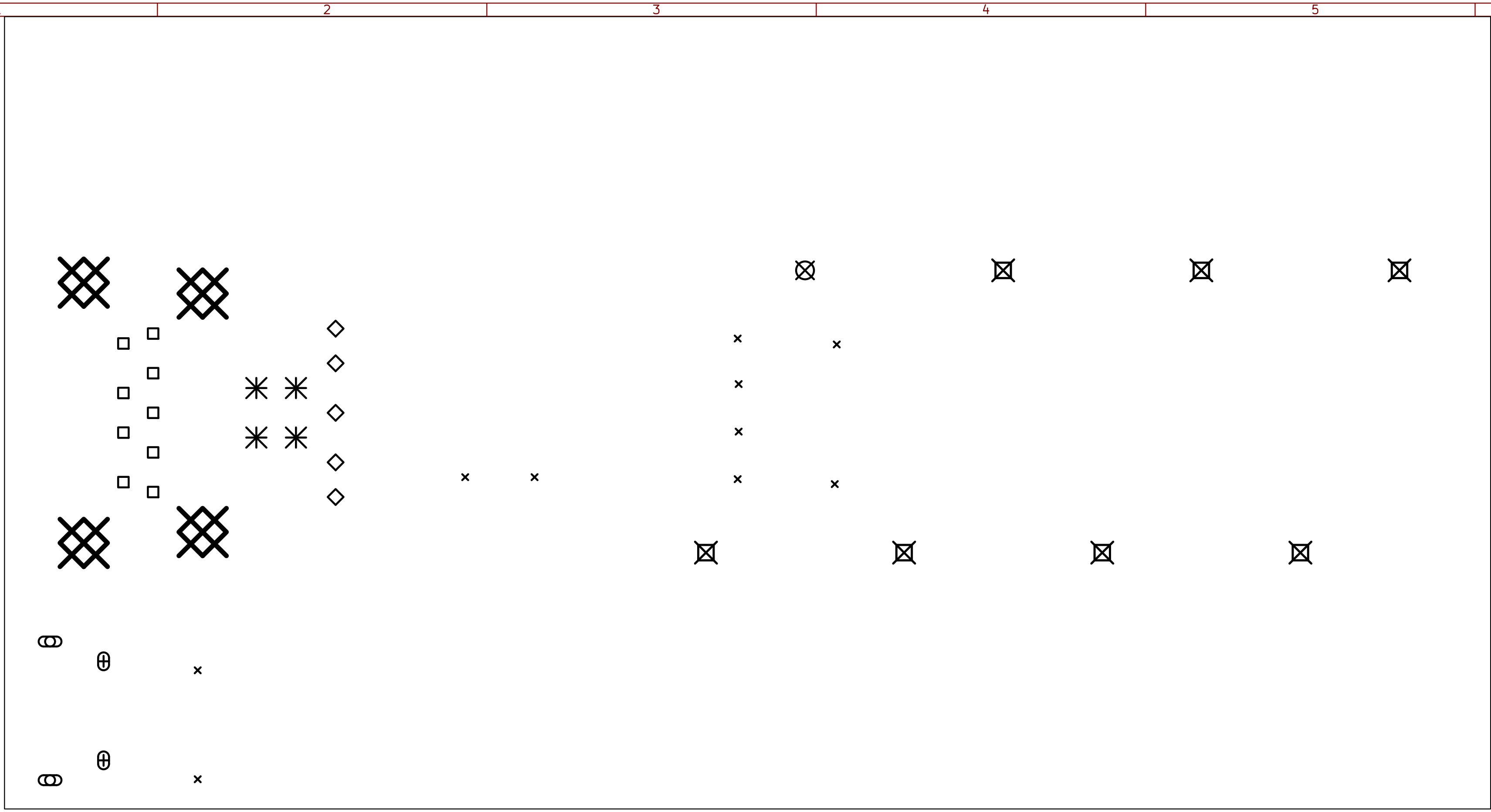

Drill Map:

0.300mm / 0.0118" (10 holes)
0.500mm / 0.0197" (0 holes + 2 slots)
0.550mm / 0.0217" (0 holes + 2 slots)
0.750mm / 0.0295" (9 holes)
0.800mm / 0.0315" (5 holes)
0.900mm / 0.0354" (1 hole)
1.020mm / 0.0402" (4 holes)
1.100mm / 0.0433" (7 holes)
2.400mm / 0.0945" (4 holes)

USB 3.0 to SFP+ Board

LAYER: Drill Map (PTH)

Repo: github.com/ringof/usb3-fiber

Designer: David Goncalves	Commit: local	OSH Licence: CERN-OHL-P-2.0
Size: A4	Date: 2026-03-13	**Rev: DEV**
	File: usb3_fiber.kicad_pcb	**Page 11 of 11**